

SYSTEM SPECIFICATION

Functional Specification

Business and system requirements for event waiver collection

System	NSNS Waiver Application
Status	As implemented
Technology	ASP.NET Core Razor Pages / .NET 10 / MySQL / Dapper
Prepared	2026-07-30

Purpose: Defines what the NSNS Waiver Application must do, the rules it enforces, the information it stores, and the acceptance conditions for operation.

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1. Purpose and scope

The application lets a customer open an event-specific link, enter contact information, add family members, review and sign a waiver, choose media-release consent, and receive confirmation. It also queues a business-owner notification and provides a protected administrator list of recent submissions.

1.1 In scope

- Public waiver form selected by the event query-string parameter.
- Server-side event validation against configuration.
- Contact details, electronic signature, family members, and optional media consent.
- Transactional MySQL persistence and email outbox creation.
- Customer confirmation and business-owner notification emails.
- Cookie-protected administrator login and paginated submission list.
- Docker-compatible deployment with Railway-oriented configuration.

1.2 Out of scope

- Self-service event management or an Events database table.
- Waiver versioning, agreement snapshots, agreement hashes, or metadata tables.
- Duplicate-submission prevention.
- Search, export, customer accounts, payment collection, or email-template administration.
- Automated legal approval of the agreement content.

2. Stakeholders and operating context

Actor	Goal	Authorized capabilities
Customer / signer	Submit a waiver for an event.	View event and agreement; enter data; add family members; sign; choose media consent; submit.
Business owner	Receive notice and review submissions.	Receive notification email; sign in; view and sort recent submissions; sign out.
Operator / maintainer	Configure and run the service.	Configure events, owner email, SMTP, admin credentials, database, and deployment.
Email worker	Deliver queued messages reliably.	Poll eligible outbox rows; send; mark sent; retry; abandon after maximum attempts.
Primary entry point: Customers use a URL such as https://waiver.example.com/?event=summer-camp-2026 . The event code is never accepted solely from the browser; it is resolved against server configuration.		

3. Functional requirements

3.1 Public event and form

ID	Requirement	Verification
FR-001	The system shall read the event code from the event query-string parameter.	Page test
FR-002	The system shall resolve the event code against configured allowed events using trimmed, case-insensitive matching.	Service test
FR-003	The system shall show an event-not-found state when the event is missing or invalid.	Page test
FR-004	The form shall collect first name, last name, optional WeChat name, email, and phone number.	UI review
FR-005	The form shall allow zero to ten family members, each with first name, last name, and optional relationship.	Page and service tests
FR-006	The system shall render the agreement from Content/waiver-agreement.html and disable submission when it is unavailable or contains the placeholder marker.	Provider/page tests
FR-007	The customer shall be required to accept the liability waiver before submission.	Page/service tests
FR-008	The customer shall enter a legal name as an electronic signature.	Validation test
FR-009	The media-release checkbox shall be optional, selected by default, and may be cleared before submission.	Page/service tests

3.2 Submission processing

ID	Requirement	Verification
FR-010	The server shall revalidate all posted values regardless of browser validation.	Service tests
FR-011	The server shall generate the signed timestamp in UTC and a UUID submission reference.	Service test
FR-012	The system shall store the configured event code and event name with each submission.	Service/repository tests
FR-013	The system shall allow duplicate submissions.	Service test
FR-014	The submission, family members, customer email, and owner email shall be created in one database transaction.	Repository test
FR-015	A successful submission shall redirect to a confirmation page showing the event, reference, and signed time.	Page test

3.3 Email delivery

ID	Requirement	Verification
FR-016	The system shall queue a professional confirmation email to the customer.	Service test

ID	Requirement	Verification
FR-017	The system shall queue an owner notification containing customer, signature, family, event, and media-consent details.	Service test
FR-018	The owner email shall link to https://waiver.nsns.ca/Admin/Submissions .	Service test
FR-019	A background worker shall process eligible outbox messages in configured batches.	Processor test
FR-020	Failed email delivery shall use exponential retry and become abandoned after the configured maximum attempts.	Processor test

3.4 Administrator area

ID	Requirement	Verification
FR-021	Anonymous users shall be redirected to the administrator login page.	Authentication configuration
FR-022	Valid configured credentials shall create an eight-hour sliding authentication cookie.	Credential/page tests
FR-023	Post-login redirects shall accept only local return URLs.	Code review
FR-024	The admin list shall show event, customer details, family members, media consent, signature, and UTC signed time.	UI/repository review
FR-025	The list shall sort only by an allow-listed set of columns.	Repository/page tests
FR-026	The list shall show at most 200 recent submissions, 20 per page, and no more than ten pages.	Page/repository tests
FR-027	The administrator shall be able to sign out and delete the authentication cookie.	Page test

4. Business and validation rules

Rule	Definition
BR-001 Event	A submission is valid only when its normalized event code exists in Waiver:Events and has a nonblank name.
BR-002 Required text	First name, last name, email, phone, and signature are trimmed and required.
BR-003 Lengths	Names and relationship: 100 characters; email: 320; phone: 40; signature: 200; IP address: 45; user agent: 500.
BR-004 Email	Email must parse as a valid address; a lower-case normalized value is stored for lookup.
BR-005 Phone	The display value is stored as entered after trimming; a digits-only normalized value must contain at least one digit.
BR-006 Family	At most ten family members are processed. Excess members are rejected.
BR-007 Agreement	Liability-waiver acceptance is mandatory; media-release consent is independent and optional.

Rule	Definition
BR-008 Media default	New forms begin with media consent selected. The database column is NOT NULL with DEFAULT TRUE.
BR-009 Duplicates	The same customer may submit multiple waivers for the same or different events.
BR-010 Time	Stored signing, creation, attempt, retry, and sent timestamps use UTC.

5. Data requirements

5.1 Waiver submissions

The waiver_submissions table is the parent record. It stores the UUID reference, event identity, customer contact information, normalized lookup values, signature, required-waiver acceptance, media-release choice, UTC signing time, request metadata, and creation time.

5.2 Family members

Each waiver_family_members row belongs to one submission through a foreign key with ON DELETE CASCADE. Family members are deliberately stored separately rather than embedded.

5.3 Email outbox

Each email_outbox row belongs to a submission and stores recipient, subject, HTML body, delivery status, attempt count, retry schedule, sent time, and a safe error summary. Deleting a submission cascades to its queued messages.

5.4 Agreement storage

The agreement remains a deployed HTML file. It is not copied into MySQL and has no version, snapshot, hash, or metadata record.

6. Non-functional requirements

Area	Requirement
Security	Use anti-forgery protection, parameterized SQL, server-side validation, HTTP-only admin cookies, local-only redirects, and fixed-time credential comparison.
Privacy	Avoid logging PII, passwords, connection strings, or SMTP secrets. Store request metadata only as required for troubleshooting.
Reliability	Persist submissions and email messages atomically. Retry email independently so SMTP failure cannot lose a waiver.

Area	Requirement
Maintainability	Keep UI in Razor Pages, business rules in services, SQL in repositories, and settings in typed options.
Performance	Limit admin reads to the 200 newest records, page at 20 rows, clamp email batch size to 1-100, and index common lookup/order fields.
Accessibility	Use semantic headings, labels, validation messages, keyboard-focusable agreement content, and status text for dynamic family-member controls.
Portability	Run on .NET 10 with MySQL in Docker-compatible and Railway-compatible environments.

7. Acceptance criteria

1. A valid configured event opens the correct form and an unknown event cannot be submitted.
2. A customer can submit with zero or up to ten family members and can submit duplicates.
3. Clearing the optional media checkbox still permits submission and stores Declined/false.
4. A successful request creates exactly one submission, the supplied family rows, and two outbox rows in one transaction.
5. The customer reaches confirmation only after persistence succeeds.
6. The email worker marks successful messages Sent and schedules or abandons failed messages according to configuration.
7. Unauthenticated users cannot view admin submissions.
8. Admin sorting cannot inject arbitrary SQL, and pagination never exposes more than 200 recent submissions.
9. Release build and automated tests complete without code errors.

8. Assumptions, exclusions, and dependencies

- A MySQL database exists and schema migrations are applied before the updated application starts.
- The operator provides valid Waiver, Email, Admin, and ConnectionStrings configuration.
- SMTP credentials permit relay from the configured sender address.
- The business owner is responsible for legal approval and encoding quality of the HTML agreement.
- Historical rows receive media_release_agreed = TRUE when migration 002 is applied because no earlier choice was recorded.
- MySQL integration tests require WAIVERAPP_TEST_MYSQL_CONNECTION; otherwise they are skipped.

Legal-content boundary: This specification describes software behavior. It does not validate whether the waiver or media-release wording is legally sufficient.